

Portfolio Class Cheat Sheet

 Portfolio(name, spec_list, uw=None)
A Portfolio is a collection of Aggregate units sharing a discretization grid; it adds diversification, capital allocation, and distortion pricing. spec_list is a list of agg specs / programs; usually built via build('port ...'). The following tables show all  methods, and fields or properties (used interchangeably). Comments elucidate the meaning of more obscure entries.

1. Specification & creation

name, agg_list (the units), n_units, unit_names, unit_names_ex, line_names, line_names_ex, line_name_pipe, line_renamer, <unit> (each unit by name, e.g. A), value_type, program, pprogram, pprogram_html, spec, spec_ex,  nice_program,  as_severity,  create_from_sample

2. Update

log2, bs, sev_calc, discretization_calc, normalize, padding, last_update, hash_rep_at_last_update,  update,  recommend_bucket,  best_bucket,  best_window,  snap,  remove_fuzz,  trim_df,  swap_density_df

3. Moments

est prefix = estimated from FFT
agg_m, agg_cv, agg_sd, agg_var, agg_skew, est_m, est_cv, est_sd, est_var, est_skew, ex ($E[X]$)

4. Statistical functions

 cdf,  sf,  pdf,  pmf,  q (VaR),  tvar,  tvar_threshold,  var,  var_dict^[1],  percentiles,  sample, sample_df,  sample_compare,  sample_density_compare

5. Validation

valid, validation_eps, describe,  explain_validation, bs_window_df, bs_description, bs_explanation, tail_class, tail_df, tail_description, tail_explanation

6. Output dataframes

density_df^[1], density, stats_df, info, independent_density_df, independent_stats_df,  unit_density,  unit_density_df,  aligned_unit_density_df,  augmented_df, augmented_dfs, figure, bounded

7. Reinsurance

reins_density_df, reins_describe, reins_stats_df (*per-unit reinsurance is set on each Aggregate*)

8. Visualization

 plot,  scatter, figure

9. Risk and pricing

 price,  price_ccoc,  price_stand_alone,  price_pentagon,  pricing_at,  pentagon_at,  pricing_bounds,  apply_distortion, distortion, distortions, distortion_df, calibration_df,  calibrate_distortion,  calibrate_distortions,  analyze_distortion,  analyze_distortions, allocation_method,  allocation_bounds,  allocation_diagnostics,  bodoff,  add_exa,  add_exa_sample,  make_comonotonic_allocations

10. Approximations

 approximate (fit a single Aggregate to the total)

11. Meta

 help,  json,  save

Notes:
[1]: matches Aggregate.
Abbreviations: m=mean, cv=coefficient of variation, sd=standard deviation, var=variance, skew(ness); VaR=value-at-risk; est=FFT estimate; exa = $E[X_i | X]$ (conditional expected allocation).